CSCI 131 — Introduction to Programming in Python
Programming Assignment 1
Input, Arithmetic, and Output

This assignment exercises `input`, type conversion, arithmetic, and `print`.

Problem 1 — Greeting

Write a program `hello.py` which asks the user for their name and greets them by name. Use the prompt `'Enter your name: '`, including the space after the colon.

Input	Output
Chuck	Hello Chuck
Ada	Hello Ada

Problem 2 — Gross Pay

Write a program `pay.py` which asks the user for a number of hours worked and a rate of pay per hour, and prints the gross pay. Use the prompts `'Enter Hours: '` and `'Enter Rate: '`, in that order.

Hours	Rate	Output
35	2.75	Pay: 96.25
40	15.50	Pay: 620.0
12.5	20	Pay: 250.0

Problem 3 — Celsius to Fahrenheit

Write a program `fahrenheit.py` which asks the user for a temperature in degrees Celsius and prints the equivalent temperature in degrees Fahrenheit. Use the prompt `'Enter a Celsius temperature: '`. The conversion is

$$F = \frac{9}{5}C + 32.$$

Celsius	Output
100	Temperature in Fahrenheit: 212.0
37	Temperature in Fahrenheit: 98.6
0	Temperature in Fahrenheit: 32.0
-40	Temperature in Fahrenheit: -40.0

Some hints

Recall that `input` always returns a *string*, even when the user types digits. Before you can do arithmetic on a value you must convert it, either in two steps or in one:

```
rate_text = input('Enter Rate: ')
rate = float(rate_text)

rate = float(input('Enter Rate: '))
```

Use `float` rather than `int` for any value that might have a decimal point. If you skip the conversion, Problem 2 will either crash with a `TypeError` or silently print two numbers glued together.

Do not worry about the number of digits after the decimal point. `620.0` is a correct answer; you are not expected to make it read `620.00`.

In Problem 3, mind the order of operations. Python evaluates `9 / 5 * c + 32` and `9 / (5 * c + 32)` differently and will not warn you which one you meant. Run your program on the values in the table above and confirm you get the output shown before you submit.

Submission

Submit all three source files on Canvas by the end of the period. Put your name in a comment at the top of each file. If a classmate helped you, put their name in the comment as well.